MIL-S-10385F
28 June 1979
SUPERSEDING
MIL-F-10385E
26 September 1974

MILITARY SPECIFICATION

SANDTRAP ASSEMBLY

(FOR PORTABLE 6-INCH PETROLEUM PIPELINE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a sandtrap assembly for use in portable pipeline pumping station manifolds.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-W-423 - Wire, Steel, Corrosion-Resisting.

FSC 4730

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:59Z
Check the source to verify that this is the current version before use.

WW-V-51	- Valve, Angle, Check, and Globe, Bronze (125, 150, and 200 Pound) Threaded End, Flange Ends, Solder Ends, and Brazed End, For Land Use.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-T-60	- Tape: Packaging, Waterproof.

MILITARY

MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-3774	- Crates, Wood; Open 12,000- and 16,000-Pound Capacity.
MIL-C-10387	- Couplings, Clamp, Pipe; with Bolts and Synthetic-Rubber Gaskets for Grooved-End Pipe and Tube.
MIL-P-10388	- Pipe Fittings: One or More Ends Grooved.
MIL-E-52798	- Enamel, Alkyd, Camouflage.

STANDARDS

FEDERAL

FED-STD-H28/7	- Screw-Thread Standards for Federal Services, Section 7, American Standard Pipe Threads (except Dryseal and Hose Coupling Types).

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MS51953	- Nipple, Pipe, Standard Steel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:59Z
Check the source to verify that this is the current version before use.

DRAWINGS

ME

D8701-1 through -3 - Sandtrap for Pipeline Pump Station Manifold.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B2.1 - Pipe Threads (except Dryseal).
B16.9 - Factory-made Wrought Steel Buttwelding Fittings.
B16.14 - Ferrous Pipe Plugs, Bushings and Locknuts with Pipe Threads.

(Application for copies should be addressed to American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN PETROLEUM INSTITUTE (API)

SPEC 5L - Specification for Line Pipe.

(Application for copies should be addressed to American Petroleum Institute, 2101 L Street, N.W. Washington, DC 20037.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Welding Qualifications, Section IX.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 53 - Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:59Z
Check the source to verify that this is the current version before use.

A 120 - Pipe, Steel, Black and Hot-Dipped, Zinc-Coated (Galvanized) Welded and Seamless, For Ordinary Service.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, 2501 N.W. Seventh Street, Miami, FL 33125.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The sandtrap assembly shall be in accordance with Drawings D8701-1 through -3 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the sandtrap assemblies. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (first produced sandtrap assembly). The contractor shall furnish one or more sandtrap assemblies as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove that his production methods will produce sandtrap assemblies that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:59Z
Check the source to verify that this is the current version before use.

3.4 Sandtrap assembly components.

3.4.1. Caps. Buttweld caps shall conform to ANSI B 16.9, wrought steel, 14-inch.

3.4.2 Couplings.

3.4.2.1 Beveled to grooved couplings. Beveled to grooved couplings shall conform to MIL-P-10388, steel, size and length as shown on drawings.

3.4.2.2 Grooved pipe clamp couplings. Grooved pipe clamp couplings shall conform to MIL-C-10387, Class B, Gasket Grade 2, Size 14-inch.

3.4.3 Gooseneck. The gooseneck shall be steel pipe conforming to ASTM A 53 or ASTM A 120, black, 1/2-inch, standard weight as shown on the drawings.

3.4.4 Nipple. The pipe nipple shall conform to MS51953-79B.

3.4.5 Pipe. The pipe shall conform to API SPEC 5L, Standard Weight, Grade B, black, size as shown on the drawings.

3.4.6 Plug. The pipe plug shall conform to ANSI B16.14, malleable iron, square head, 2-inch.

3.4.7 Screen. The screen shall be wire conforming to QQ-W-423, Form I, Composition 304, Condition B, dimensions as shown on the drawing.

3.4.8 Valve. The valve shall conform to WW-V-51, Type I, Class A, End Connection 1, 1/2-inch.

3.5 Hydrostatic pressure. The sandtrap assembly shall withstand a hydrostatic pressure of 950 pounds per square inch (psi) without evidence of leakage, rupture, or yielding.

3.6 Threads. All threads shall be American National Taper Pipe Threads (NPT) conforming to FED-STD-H28/7 or ANSI B2.1.

3.7 Identification marking. The sandtrap assembly shall be identified in accordance with MIL-STD-130.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:59Z
Check the source to verify that this is the current version before use.

3.8 Treatment and Painting. The exterior surfaces of the sandtrap assembly shall be cleaned, treated and painted in accordance with MIL-T-704, Type A. The finish coat shall be forest green enamel conforming to MIL-E-52798.

3.9 Workmanship. The sandtrap assembly shall be free from defects such as misalined components, incomplete welds, cracks, and any other defect that could impair the function and serviceability characteristics of the sandtrap assembly.

3.9.1 Welders. Before assigning any welder to manual welding work covered by this specification the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1, Structural Welding Code, Section 5, Qualification.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS or ASME codes.

3.9.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the sandtrap assembly to conform to the examination and test requirements specified in Section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 0.188 inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the sandtrap assembly shall be at the option of the contractor.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:59Z
Check the source to verify that this is the current version before use.

facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.1.2 Disassembly inspection. Failure of any examination or test by the first produced sandtrap assembly shall be cause for disassembly, in the presence of a Government representative, of the first produced sandtrap assembly to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first produced sandtrap assembly. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

(a) First produced sandtrap assembly inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First produced sandtrap assembly inspection.

4.3.1 Examination. The sandtrap assembly shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The sandtrap assembly shall be tested as specified in 4.5.2. Failure of the test shall be cause for performing the disassembly inspection specified in 4.1.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:59Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105.

4.4.2 Examination.

4.4.2.1 Samples. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 1.0 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.3 Tests.

4.4.3.1 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. AQL shall be 1.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The sandtrap assembly shall be examined as specified herein for the following defects:

Major

101. Any dimensions not in accordance with the drawings.
102. Material not as specified.
103. Components incorrect or missing.
104. Workmanship not as specified.

Minor

201. Identification marking missing, incomplete, or illegible.
202. Treatment and painting not as specified.

4.5.2 Hydrostatic test. Subject the complete sandtrap assembly to a hydrostatic pressure of 950 psi for not less than 30 seconds. Examine the sandtrap assembly for leakage, rupture, and yielding. Nonconformance to 3.10 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:59Z
Check the source to verify that this is the current version before use.

4.6 Inspection of packaging.

4.6.1 Preproduction pack inspection.

4.6.1.1 Examination. The preproduction pack shall be examined for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

4.6.1.2 Test. The preproduction pack of the sandtrap assemblies for Level A shall be subjected to the railroad car test as specified for guided-impact tests in accordance with MIL-STD-1186, Appendix A. The car shall strike a string of five empty cars, with draft gear extended and the brakes set, at a speed of not less than 10 miles or more than 11 miles per hour. At the conclusion of the preproduction pack test, the pack shall be examined. Any shifting of contents; loosening or breaking of holddowns, ties, stays, blocking, or bracing; or any visual damage to the sandtraps shall constitute failure of the test and shall be cause for rejection.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

105.	Materials, methods, and containers not as specified for Level A. Each incorrect material, method, or container shall constitute one defect.
106.	Valve openings not sealed as specified for Level A.
107.	Components not packaged and cushioned as specified for Level A.
108.	Threaded opening of sandtrap not closed as specified for Level A.
109.	Inlet and outlet openings and land sealing surfaces not wrapped as specified for Level A.
110.	Packing not as specified for Level A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:59Z
Check the source to verify that this is the current version before use.

111. Blocking, bracing, anchoring, and shroud not in accordance with the referenced documents as specified for Level A.
112. Marking illegible, incomplete, incorrect, or missing for Level A or Commercial.

5. PACKAGING

5.1 Preproduction pack. When specified (see 6.2), the contractor shall furnish a preproduction pack for examination and test within the time frame specified (see 6.2) to prove prior to starting production packaging and packing that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination and tests shall be those specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The preproduction pack may be prepared utilizing either the first produced sandtrap assembly or a production sandtrap assembly. When the first produced sandtrap assembly is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government, to facilitate comparison between the first produced sandtrap assemblies and production sandtrap assemblies.

5.2 Preservation. Preservation shall be Level A or Commercial as specified (see 6.2).

5.2.1 Level A. The gooseneck, nipple, and valve assembly shall be removed from the sandtrap assembly. The openings in the valve assembly shall be sealed with tape conforming to PPP-T-60, Type IV and placed together with gooseneck and nipple in a fiberboard box conforming to PPP-B-636, Class Weather Resistant. The box shall be closed in accordance with the appendix to the box specification, Method V. The contents shall be cushioned to prevent damage. The box shall be secured inside the inlet end of the sandtrap with tape conforming to PPP-T-60, Type IV. The threaded openings in the sandtrap shall be closed with plastic cap plugs or tape conforming to PPP-T-60, Type IV. The inlet and outlet opening of the sandtrap shall be covered with barrier material conforming to MIL-B-121, Type I, Grade A. The barrier shall extend back beyond the groove a minimum distance of 0.25 inch. The wrap shall be secured with tape conforming to PPP-T-60, Type IV. The tape shall be a width that will completely cover the land sealing surface and extend back beyond the edge of the barrier material not less than 0.50 inch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:59Z
Check the source to verify that this is the current version before use.

5.2.2 Commercial. The sandtrap assemblies shall be preserved in accordance with MIL-STD-1188.

5.3 Packing. The sandtrap assemblies shall be packed, Level A or Commercial as specified (see 6.2).

5.3.1 Level A. Two or more sandtrap assemblies, as specified (see 6.2), shall be packed together in an open wood crate conforming to MIL-C-3774, Type I, Style A. The contents shall be blocked, braced, anchored, and protected with a full shroud in accordance with the appendix to the crate specification and MIL-STD-1186.

5.3.2 Commercial. The sandtrap assemblies shall be packed in accordance with MIL-STD-1188.

5.4 Marking.

5.4.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.4.2 Commercial. Marking for Commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The sandtrap assembly is intended for use as a component of a pumping station manifold in a pipeline system conveying light petroleum products.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of first produced model, and number of sandtrap assemblies required (see 3.2).
(c) When the welding process will not be at the option of the contractor (see 3.14.2).
(d) When preproduction pack is required (see 5.1).
(e) Time frame required for submission of preproduction pack (see 5.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:59Z
Check the source to verify that this is the current version before use.

(f) Degree of preservation and degree of packing required (see 5.2 and 5.3).
(g) Quantity to be packed (see 5.3.1).

6.3 First produced sandtrap assembly. Any changes or deviations in production sandtrap assemblies from the approved first produced sandtrap assembly during production will be subject to the approval of the contracting officer. Approval of the first produced sandtrap assembly will not relieve the contractor of his obligation to furnish sandtrap assemblies conforming to this specification.

6.4 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, pack, and mark the sandtrap assemblies in accordance with this specification.

6.5 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

Custodians:
Army - ME
Air Force - 99

Review activity:
DLA - CS

Preparing activity:
Army - ME

Project 4730-0941

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:59Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product *are invited to submit comments* to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-S-10385F Sandtrap Assembly (for Portable 6-Inch Petroleum Pipeline)

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:59Z
Check the source to verify that this is the current version before use.